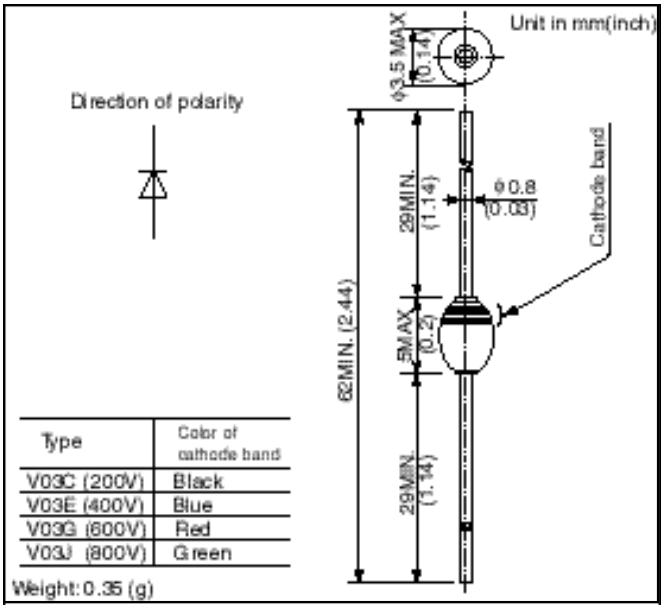
V03

FEATURES

- For general purpose.
- Diffused-junction. Glass passivated and encapsulated.

OUTLINE DRAWING



ABSOLUTE MAXIMUM RATINGS

Item	Type		V03C	V03E	V03G	V03J
Repetitive Peak Reverse Voltage	V_{RRM}	V	200	400	600	800
Non-Repetitive Peak Reverse Voltage	V_{RSM}	V	300	500	800	1000
Average Forward Current	$I_{F(AV)}$	A	1.3 (Single-phase half sine wave 180° conduction, TL = 90°C, Lead length = 10mm)			
Surge(Non-Repetitive) Forward Current	I_{FSM}	A	40 (Without PIV, 10ms conduction, Tj = 175°C start)			
I^2t Limit Value	I^2t	A^2s	6.4 (Time = 2 ~ 10ms, I = RMS value)			
Operating Junction Temperature	T_j	°C	-65 ~ +175			
Storage Temperature	T_{stg}	°C	-65 ~ +200			

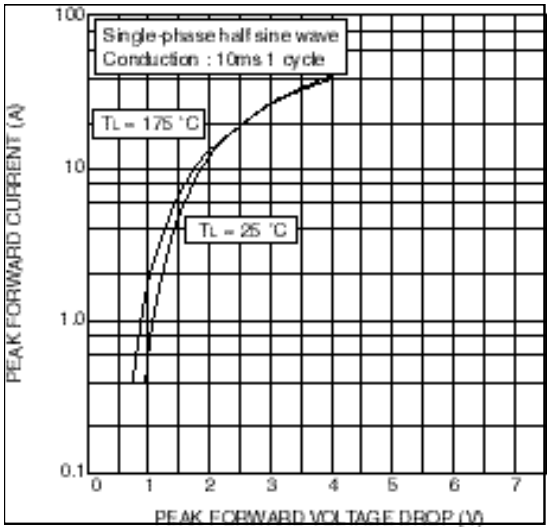
Notes (1) Lead mounting : Lead temperature 300°C max. to 3.2mm from body for 5sec. max..
(2) Mechanical strength : Bending 90°×2 cycles or 180°×1 cycle, Tensile 2kg, Twist 90°×1 cycle.

CHARACTERISTICS(T_L=25°C)

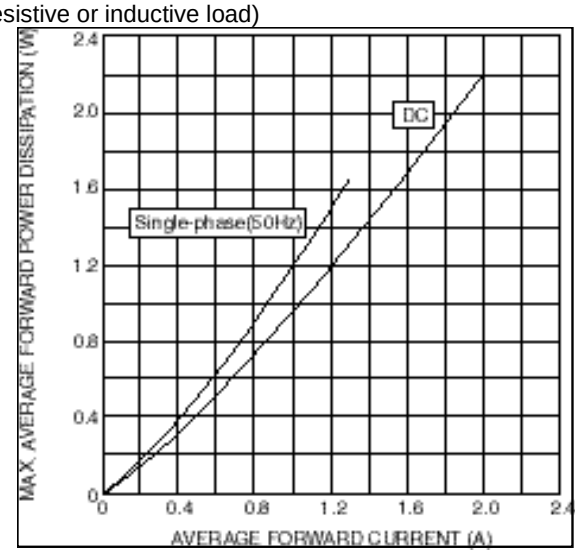
Item	Symbols	Units	Min.	Typ.	Max.	Test Conditions	
Peak Reverse Current	I_{RRM}	μA	—	1.5	20	C class	Rated V_{RRM}
				0.6	10	E,G,J class	
Peak Forward Voltage	V_{FM}	V	—	—	1.1	I_{FM}=1.3Ap, Single-phase half sine wave 1 cycle	
Reverse Recovery Time	trr	μs	—	3.0	—	I_F=2mA, V_R=-15V	
Steady State Thermal Impedance	$R_{th(j-a)}$	°C/W	—	—	80	Lead length = 10 mm	
	$R_{th(j-l)}$				50		

V03

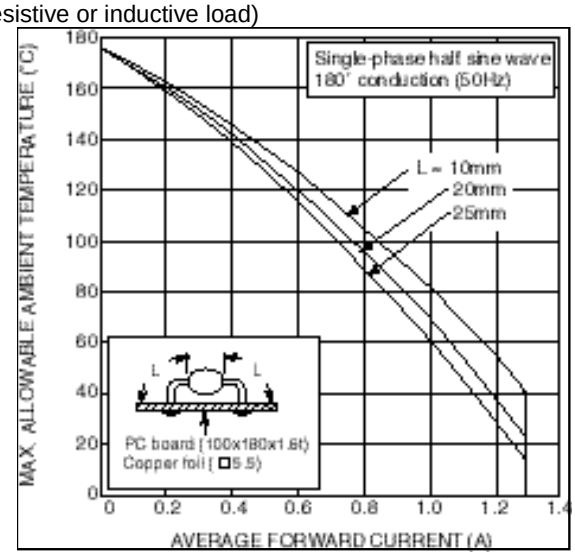
Forward characteristics



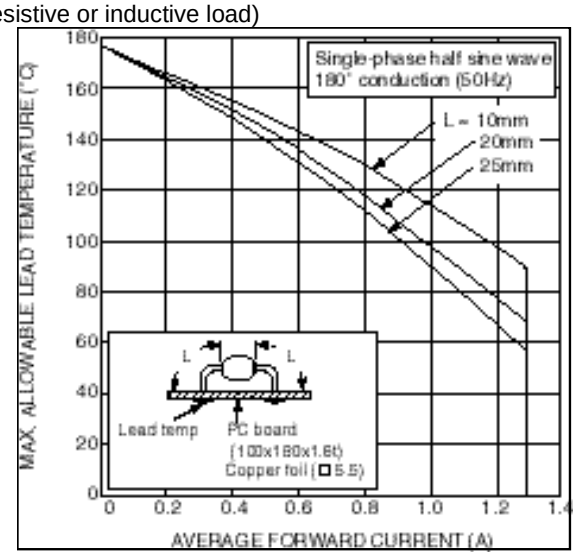
Max. average forward power dissipation
(Resistive or inductive load)



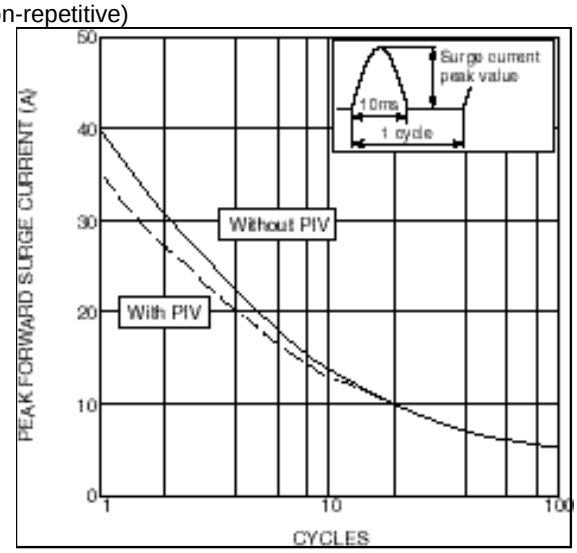
Max. allowable ambient temperature
(Resistive or inductive load)



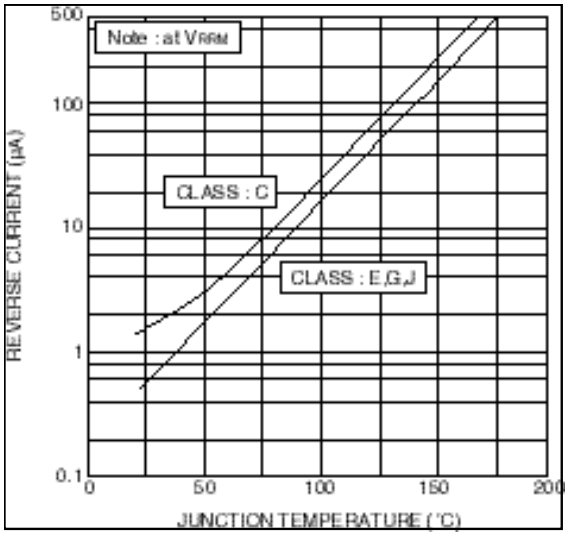
Max. allowable lead temperature
(Resistive or inductive load)



Surge forward current characteristics
(Non-repetitive)

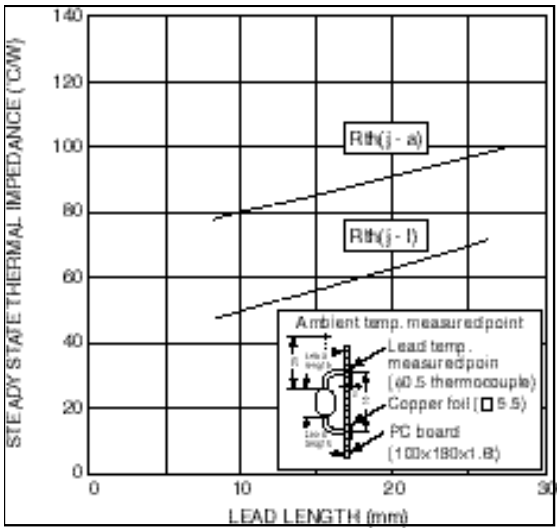


Typ. reverse current vs. junction temperature

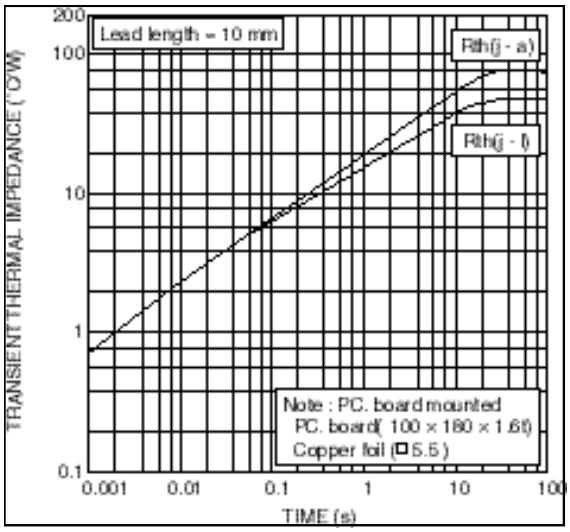


V03

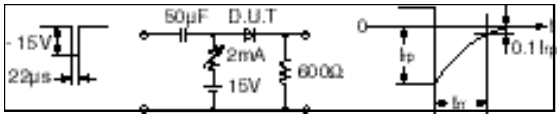
Steady state thermal impedance



Transient thermal impedance



Reverse recovery time(trr) test circuit



Precautions for Safe Use and Notices

If semiconductor devices are handled inappropriate manner, failures may result.
For this reason, be sure to read “Precaution for Use” before use.

⚠ This mark indicates an item about which caution is required.

⚠ **CAUTION** This mark indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury and damage to property.

⚠ CAUTION

(1) Regardless of changes in external conditions during use “absolute maximum ratings” should never be exceed in designing electronic circuits that employ semiconductors.In the case of pulse use, furthermore,″safe operating area(SOA)”precautions should be observed.

(2) Semiconductor devices may experience failures due to accident or unexpected surge voltages. Accordingly, adopt safe design features, such as redundancy or prevention of erroneous action, to avoid extensive damage in the event of a failure.

(3) In cases where extremely high reliability is required (such as use in nuclear power control, aerospace and aviation, traffic equipment, life-support-related medical equipment, fuel control equipment and various kinds of safety equipment), safety should be ensured by using semiconductor devices that feature assured safety or by means of user’s fail-safe precautions or other arrangement. Or consult Hitachi’s sales department staff.

(If a semiconductor device fails, there may be cases in which the semiconductor device, wiring or wiring pattern will emit smoke or cause a fire or in which the semiconductor device will burst)

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http://www.hitachi-power-semiconductor-device.co.jp/en/